

Automated Regression Analysis & Reporting Pipeline

Overview

This project automates the end-to-end processing of QA regression test results that are shared via unstructured emails. The solution converts raw regression emails into **structured, comparable, and decision-ready insights** for Business Analysts (BAs) and Product Managers (PMs).

Instead of manually analyzing regression results across multiple modules and cycles, stakeholders receive a **single consolidated summary** highlighting regressions, improvements, and unchanged areas.

Problem Statement

QA teams share regression results via large text-based emails containing module-wise metrics such as:

- Expected test cases
- Passed tests
- Failed tests
- Executed tests
- Unexecuted tests

Challenges

- Manual comparison with previous cycles
- High risk of missed regressions
- Time-consuming analysis across ~90 modules
- No centralized historical record
- Subjective interpretation of result

Solution

I designed and implemented an automated workflow that:

- Listens for incoming QA regression emails
- Extracts cycle and module-level metrics
- Compares current results with the previous cycle
- Calculates deltas and identifies regressions
- Classifies each module with a clear status
- Stores historical data centrally
- Sends a clean, consolidated summary email to stakeholders

The entire process runs without any manual intervention.

End-to-End Workflow

QA Regression Email → Email Parsing & Normalization → Cycle Detection → Module-wise → Metric Extraction → Historical Comparison → Delta Calculation → Status Classification → Central Storage (SharePoint) → HTML Summary Email to BA / PM

Key Logic & Design Decisions

1. Cycle Management

- Extracts regression date from email subject
- Automatically determines the next cycle number
- Handles first-run and missing historical data safely

2. Metric Extraction

For each module:

- Expected
- Passed
- Failed
- Executed

- Unexecuted

All values are normalized into integers for accurate comparison.

3. Delta Analysis

Each module is compared with its previous cycle:

- Δ Failed
- Δ Executed
- Δ Unexecuted

4. Status Classification

Business-friendly classification logic:

Condition	Status
No change in deltas	NO CHANGE
Failed tests decreased	IMPROVED
Failed tests increased	REGRESSION FAILS

This removes the need for stakeholders to interpret raw numbers.

Output for Stakeholders

1. Centralized Historical Record

- Module-wise, cycle-wise data stored centrally
- Enables audits, trend analysis, and dashboards

2. Automated Summary Email

- Single HTML table
- One row per module
- Clear deltas and status indicators
- Ready for decision-making

Example columns:

- Module
- Δ Executed

- Δ Failed
- Δ Unexecuted
- Status

Tech Stack

- Power Automate – workflow orchestration
- SharePoint Lists – system of record & history
- Expression-based parsing & validation
- HTML generation for reporting

Impact

- Reduced manual analysis effort to zero
- Faster and more reliable release decisions
- Eliminated subjective interpretation of regression results
- Scales across ~90 modules per cycle
- Fully auditable and reusable

Future Enhancements

- Regression trend dashboards
- Threshold-based alerts for critical failures
- Integration with release decision workflows
- API-based input instead of email

Input Email

Subject: vlmazOmniApp001 moddev CART result 20260314

System:moddev Object:LN2 20260314

```
*****
*** EXPECTED NUMBER OF TEST CASES:  237,122
*** Number of Passed Tests:         237,097
*** Number of Failed Tests:          25
*** Total Number of Tests Executed:  237,122
*** Number of Unexecuted Tests:      0
*****
```

Output Email

Hello Team,

Please find below the full summary of all modules:

Module	Δ Executed	Δ Failed	Δ Unexecuted	Status
OCSO	0	1	0	REGRESSION FAILS
SP	0	1	0	REGRESSION FAILS
ST	0	0	0	NO CHANGE
TA	0	-1	0	IMPROVED
TX	0	0	0	NO CHANGE
UT	0	0	0	NO CHANGE
WH	0	0	0	NO CHANGE
WHPX	0	1	0	REGRESSION FAILS
DS	0	0	0	NO CHANGE

Database (SharePoint List)

Regression Result Records

Regression Result ☆ ☺

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Module Name	Cycle Date ☹	CycleNumber	Expected	Passed	Failed	Executed	Unexecuted	ComparedToPrevious	DifferencesJSON
OCSO	3/12/2026	6	17,061	170,961	0	170,961	0	true	Failed changed: 11 → 0; Passed changed: 170950 → 170961;
SP	3/12/2026	6	3,142	3,142	0	3,142	0	false	
ST	3/12/2026	6	21,081	21,080	1	21,081	0	false	
TA	3/12/2026	6	20,599	20,205	394	20,599	0	false	
TX	3/12/2026	6	1,467	1,467	1	1,468	1	false	
UT	3/12/2026	6	92	91	1	92	0	false	
WH	3/12/2026	6	58,227	57,662	565	58,227	0	false	
WHPX	3/12/2026	6	34,291	34,288	3	34,291	0	false	
DS	3/12/2026	6	90	89	1	90	0	false	

Regression Summary Records

Regression Summary ☆ ☺							
Status:		IMPROVED ✕	REGRESSION FAILS ✕				
CycleNumber	Module	DeltaFailed	DeltaExecuted	DeltaUnexecuted	Status ⚙	CycleStatus	
12	FEPX	-1	0	0	IMPROVED		
12	GLVR	-1	0	0	IMPROVED		
12	PE	-1	0	0	IMPROVED		
12	PL	-2	0	0	IMPROVED		
13	FEPX	1	0	0	REGRESSION FAILS	Inprogress	
13	GLVR	1	0	0	REGRESSION FAILS	Inprogress	
13	PE	1	0	0	REGRESSION FAILS	Inprogress	
13	PL	2	0	0	REGRESSION FAILS	Inprogress	
13	DNVS	-2	0	0	IMPROVED	Inprogress	
14	TA	1	0	0	REGRESSION FAILS		